

Retail Spatial Intelligence

CCTV Foot-Traffic & Spatial Behavior Report

VERIFIED COMPUTER VISION DATA

Analysis Parameters & Provenance

VIDEO SOURCE:	sample_retail_cctv.mp4	DATE PROCESSED:	Oct 6, 2026, 10:03 PM
RESOLUTION:	768x432 (768x432)	VIDEO DURATION:	49.67s (596 frames @ 12 fps)
ZONE CONFIG:	Automatic Camera Regions	ANALYSIS QUALITY:	High Quality
CV PIPELINE:	YOLOv8 + ByteTrack (Floor Footpoints)	REPORT VERSION:	v2.2 (Canonical Single Truth)
TRACKING QUALITY:	Good (7 valid tracks)	JOB AUDIT ID:	f2c5b1a5-2c62-4bab

TRACKED VISITORS

7

Unique valid tracks

AVG ZONE DWELL

2.53s

Median: 2.5s

PEAK CONCURRENCY

3

Max simultaneous

TOTAL ZONE VISITS

11

Across 4 zones

Executive Summary

DETERMINISTIC SPATIAL RULES SUMMARY

Video analysis observed 7 unique tracked customer visits across 49.67s of footage. Overall average dwell time was 2.53s with a peak simultaneous store occupancy of 3 people. 2 potential bottleneck area(s) and 1 potential dead-zone area(s) were deterministically identified.

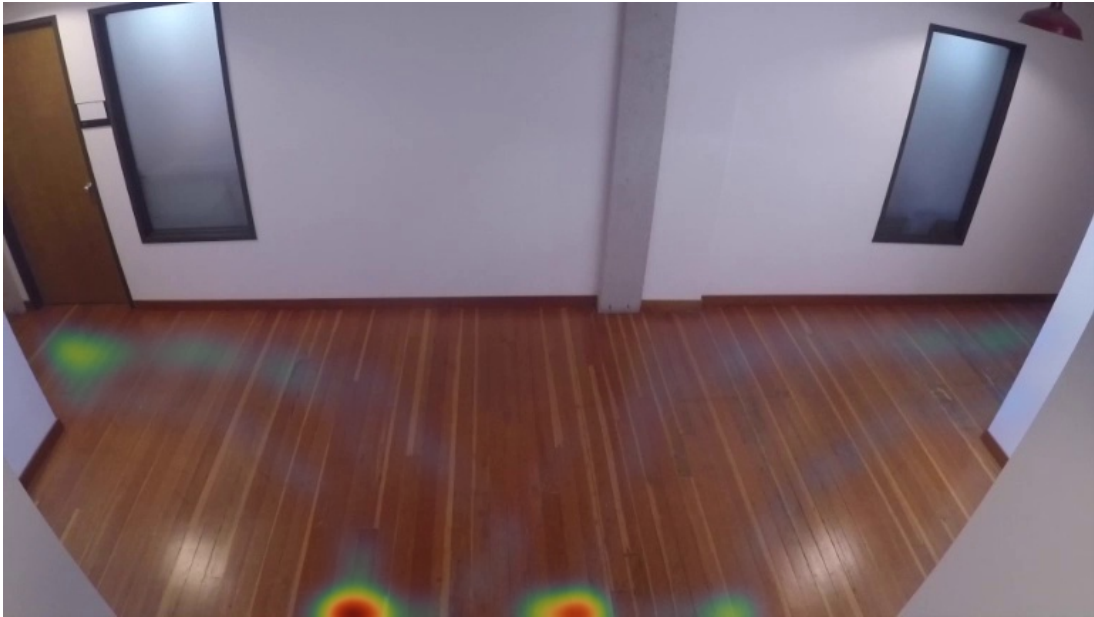
1. Temporal Traffic Dynamics & Concurrency

Primary traffic flow centers around Region 3 (Lower Left Sector). Most common movement pathway: Region 3 (Lower Left Sector). Observed 6 distinct zone-to-zone transitions.

PEAK STORE OCCUPANCY:	3 concurrent tracked individuals
AVERAGE OCCUPANCY:	0.65 people per analyzed interval
IMAGE-SPACE SPEED:	99 px/s (Relative pedestrian velocity)

2. 2D Gaussian Kernel Spatial Heatmap

Accumulated ground foot-points smoothed via 2D Gaussian density kernel and rendered onto the representative camera frame:



Heatmap represents observed customer foot-point activity during the analyzed period. Red indicates peak dwell density.

3. Spatial Zone Performance Analysis (Normalized)

Comprehensive empirical measurements per spatial zone. Normalized area metrics prevent large zones from appearing artificially dominant.

Zone Name	Visits	Unique	Avg Dwell	Max Dwell	Peak	Area (%)	Density	Vis/Min	Share
Region 1 (Upper Left Sector)	0	1	0s	0s	1	22%	9.59	1.21	0%
Region 2 (Upper Right Sector)	2	2	1.5s	2.5s	1	22%	28.77	2.42	18.2%
Region 3 (Lower Left Sector)	5	5	3.1s	4.83s	2	22%	138.36	6.04	45.5%
Region 4 (Lower Right Sector)	4	5	2.33s	4.66s	2	22%	87.67	6.04	36.4%

4. Customer Flow & Traversal Pathways

Zone-to-zone transitions mined from consecutive spatial crossings across verified customer paths:

DETECTED DIRECT ZONE TRANSITIONS:

- Region 4 (Lower Right Sector) !' Region 2 (Upper Right Sector) : 3 visitor(s) (25%)
- Region 3 (Lower Left Sector) !' Region 1 (Upper Left Sector) : 3 visitor(s) (25%)
- Region 4 (Lower Right Sector) !' Region 3 (Lower Left Sector) : 2 visitor(s) (16.7%)
- Region 1 (Upper Left Sector) !' Region 3 (Lower Left Sector) : 2 visitor(s) (16.7%)
- Region 3 (Lower Left Sector) !' Region 4 (Lower Right Sector) : 1 visitor(s) (8.3%)

Frequent Customer Journey Sequences:

1. Region 3 (Lower Left Sector)
Observed in 2 customer(s) (28.6% of journeys) - Avg Duration: 4.7s
2. Region 3 (Lower Left Sector) !' Region 4 (Lower Right Sector) !' Region 2 (Upper Right Sector)
Observed in 1 customer(s) (14.3% of journeys) - Avg Duration: 4.3s
3. Region 4 (Lower Right Sector) !' Region 3 (Lower Left Sector) !' Region 1 (Upper Left Sector) !' Region 3 (Lower Left Sector) !' Region 1 (Upper Left Sector)
Observed in 1 customer(s) (14.3% of journeys) - Avg Duration: 5.5s
4. Region 4 (Lower Right Sector)
Observed in 1 customer(s) (14.3% of journeys) - Avg Duration: 4.7s

5. Forensic Bottleneck & Dead-Zone Intelligence

POTENTIAL BOTTLENECK ANALYSIS:

- **Region 3 (Lower Left Sector) (Bottleneck Score: 80.2/100 - Confidence: Medium)**
 - Evidence: 67% relative store peak occupancy (2 simultaneous customer tracks)
 - Evidence: Prolonged dwell duration: 3.1s exceeds store average (2.53s)
 - Evidence: Elevated spatial traffic density: 6.85 visitors per 100k sq px
- **Region 4 (Lower Right Sector) (Bottleneck Score: 72.5/100 - Confidence: Medium)**
 - Evidence: 67% relative store peak occupancy (2 simultaneous customer tracks)
 - Evidence: Unconstrained movement speed: 104.5 px/s indicates fluid traversal
 - Evidence: Elevated spatial traffic density: 6.85 visitors per 100k sq px

POTENTIAL DEAD-ZONE ANALYSIS:

- **Region 1 (Upper Left Sector) (Deficiency Score: 90/100, Activity: 10% - Confidence: High)**
 - Evidence: Zero customer visits recorded during observed footage
 - Evidence: Low customer penetration: only 14.3% of tracked individuals visited (1 visitors)
 - Evidence: Negligible concurrency: maximum 1 customer present simultaneously

HIGH-ACTIVITY DESTINATION ZONES:

- Region 3 (Lower Left Sector): 5 visits, 3.1s avg dwell, peak concurrent: 2 (High Activity Corridor)
- Region 4 (Lower Right Sector): 4 visits, 2.33s avg dwell, peak concurrent: 2 (High Activity Corridor)

6. Strategic Recommendations & Priority Action Plan

[Congestion Alleviation] Region 3 (Lower Left Sector)

Evidence: 67% relative store peak occupancy (2 simultaneous customer tracks); Prolonged dwell duration: 3.1s exceeds store average (2.53s); Elevated spatial traffic density: 6.85 visitors per 100k sq px
Action: Widen primary aisle clearance or reposition high-interaction displays in Region 3 (Lower Left Sector) to alleviate observed transit bottlenecks.

[Foot-Traffic Revitalization] Region 1 (Upper Left Sector)

Evidence: Zero customer visits recorded during observed footage; Low customer penetration: only 14.3% of tracked individuals visited (1 visitors); Negligible concurrency: maximum 1 customer present simultaneously
Action: Introduce focal visual merchandising, promotional signage, or destination product categories in Region 1 (Upper Left Sector) to redistribute foot-traffic from congested zones.

Prioritized Action Plan:

- [High Priority]:** Audit aisle clearance in Region 3 (Lower Left Sector), Region 4 (Lower Right Sector) to eliminate movement slowdowns.
- [Medium Priority]:** Review product placement and category visibility in Region 1 (Upper Left Sector).
- [Low Priority]:** Conduct subsequent video analysis after implementing spatial adjustments to evaluate before/after metrics.

7. Analytics Validation & Ground-Truth Integrity

DETECTION VALIDATION:
193 raw detections | 0.83 avg conf | 0.32 det/frame

TRACKING VALIDATION:
7 valid tracks (0 transient filtered) | Max duration: 5.5s | ~0 ID switch candidate(s)

Epistemological Scope: What The System Actually Knows

THE SYSTEM CAN DIRECTLY OBSERVE:

- Spatial pedestrian movement & footpaths
- Zone entry/exit timestamps & dwell duration
- Concurrent physical occupancy & density
- Sequential zone transition flow & friction

THE SYSTEM CANNOT DIRECTLY OBSERVE:

- Customer identity or demographic PII
- Product purchase decisions or basket value
- Revenue or sales conversion (requires POS)
- Real-world metric distance without 3D calibration

Scientific Methodology & Limitations

- Single-Camera Monocular Perspective: Field of view is restricted to camera mounting angle. Speeds are reported in image-space pixels/second, not calibrated metric coordinates.
- Visual Occlusion: Dense clusters or shelving can temporarily occlude foot points. The platform uses a tolerance buffer, but prolonged occlusions may fragment tracks.
- Zero PII Policy: The system operates with transient anonymous IDs only. No face recognition, demographic profiling, or biometric storage is performed.
- Automatic Regions: Partitioned geometrically across the camera field of view; they represent spatial camera sectors, not semantic store departments.

AUTHENTICATION & VERIFICATION STAMP

Generated deterministically by Retail Spatial Intelligence Platform Engine v2.2. All metrics directly traceable to video frame detections. Job ID: f2c5b1a5-2c62-4bab-8d03-687cab8094b6

